

Corso: Large Language Models
Quiz: Exam (2025-01-29)

Iniziato	29 gennaio 2025, 17:11
Stato	Completato
Terminato	29 gennaio 2025, 18:26
Tempo impiegato	1 ora 15 min.
Valutazione	15,50 su un massimo di 18,00 (86%)
Riepilogo del tentativo	1 2 3 4 5 6 7 8 9 10 11 12

Domanda 1

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-25% penalty for each wrong answer)

Which of the following are benefits of a Retrieval-Augmented-Generation approach to prompting?

☐

 a. It is possible to adopt self-critique mechanisms to refine the outputs of the agent.

☒

 b. It is possible to reduce the amount of false information provided to the final user. ✓

☒

 c. It is possible to prevent the obsolescence of answers by using up-to-date documents. ✓

☐

 d. It is possible to reduce the amount of hallucinations by providing a fixed structure for the agent’s answers.

☐

 e. It is possible to decompose a task in several smaller steps.

Risposta corretta.

Le risposte corrette sono:

It is possible to reduce the amount of false information provided to the final user.,

It is possible to prevent the obsolescence of answers by using up-to-date documents.

Domanda 2

Risposta
errata

Punteggio
ottenuto -0,25
su 1,00

1 point (-25% penalty for each wrong answer)

In-context Learning has been a fundamental paradigm shift for Language Models because:

☐

 a. It replaces the need for attention mechanisms by introducing a sequential input processing

☐

 b. It improves the efficiency of training by compressing longer sequences into fixed-size representations

☒

 c. None of the other options is correct ✗

☐

 d. It focuses on training models to explicitly output the reasoning steps required for solving a task, instead of directly giving the answer

☐

 e. It introduces the ability to train models with fewer data points by leveraging pre-trained embeddings

☐

 f. It allows models to perform tasks by conditioning on examples provided in the input, without requiring task-specific fine-tuning

☐

 g. It enables models to store task-specific knowledge directly in their architecture, removing the need for example inputs

Your answer is incorrect.

La risposta corretta è: It allows models to perform tasks by conditioning on examples provided in the input, without requiring task-specific fine-tuning

Domanda 3

Completo

Punteggio
ottenuto 2,00
su 2,00**2 points (no penalty for a wrong answer)****Consider the following function**

```
def calculate_total_cost(items, discount, tax_rate, coupon_code, membership_status):  
    total_cost = sum(items)  
    discounted_cost = total_cost * (1 - discount / 100)  
    if coupon_code == "SUMMER21":  
        discounted_cost *= 0.8 # Apply an additional 20% discount for the coupon  
    final_cost_with_tax = discounted_cost * (1 + tax_rate / 100)  
    if membership_status:  
        final_cost_with_tax *= 0.9 # Premium members get an additional 10% off  
    if final_cost_with_tax < 0:  
        final_cost_with_tax = 0  
    return final_cost_with_tax
```

An LLM engine generates the following test cases

- **T1:** assert calculate_total_cost([100, 50], 10, 5, "SUMMER21", True) == 102.06
- **T2:** assert calculate_total_cost([200, 100], 20, 10, "WINTER21", False) == 264
- **T3:** assert calculate_total_cost([50], 0, 0, "", False) == 50.0
- **T4:** assert calculate_total_cost([120, 80], 15, 5, "SUMMER21", False) == 162.8

a) What is the pass rate of the test suite against the correct code?

b) Considering the following mutations:

- Line 3: discounted_cost = total_cost * (1 - discount / 10)
- Line 6: discounted_cost * (1 - tax_rate / 100)
- Line 9: if final_cost_with_tax == 0:

What is the mutation score of the test suite?

[Note: discuss and justify the results if needed]

Pass rate: All tests pass except for T4 where the result is 142.8 against the expected 162.8
Pass rate = 75.0%

Mutation score:

T1 is able to kill the first mutant.

T1 is able to kill also the second mutation (92.34 against expected 102.06)

None of the tests run line 9 so the third mutation is impossible to kill.

Mutants killed: 2/3 -> 66.7%

Commento:

Domanda 4

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-25% penalty for each wrong answer)

Adapter layers are introduced to reduce the number of tunable parameters. They consist in:

- ☐ a. A task-specific head added to the model, which is fine-tuned while the rest of the model is kept frozen
- ☐ b. A method for splitting a large model into smaller sub-models that are trained independently and later combined
- ☐ c. Replacing the feedforward layers in Transformers with low-rank matrix decompositions to reduce memory usage
- ☒ d. Small, trainable layers inserted into a pre-trained model, which are fine-tuned while the rest of the model is frozen ✓
- ☐ e. A technique to merge multiple pre-trained models into a single model with fewer parameters
- ☐ f. Additional layers that replace the attention mechanism in the Transformer architecture to simplify computation
- ☐ g. None of the others is a valid description of adapters

Your answer is correct.

La risposta corretta è:

Small, trainable layers inserted into a pre-trained model, which are fine-tuned while the rest of the model is frozen

Domanda 5

Parzialmente
corretta

Punteggio
ottenuto 0,75
su 1,00

1 point (-25% penalty for each wrong answer)

Which of the following is/are **not** a technique to evaluate the quality of test cases generated with LLMs?

- ☐ a. Execution Success Rate
- ☐ b. Coverage
- ☐ c. Mutation score
- ☒ d. Functional Correctness ✓
- ☒ e. Precision and recall ✗

Risposta parzialmente esatta.

Hai selezionato troppe opzioni.

La risposta corretta è:

Functional Correctness

Domanda 6Risposta
correttaPunteggio
ottenuto 2,00
su 2,00**2 points (no penalty for a wrong answer)**

For the following encoder-only transformer model, compute:

1. the number of parameters for the embedding layer (token + positional)
2. the number of parameters of the transformer (excluding the embedding layer, or head layer(s))

Note: assume that none of the layers uses a bias term. You can ignore layer normalization parameters.

- Vocabulary size $V=50,265$
- Embedding/model dimension $d=1024$
- Positional embeddings: absolute, learned, for 514 positions
- Number of layers $L=24$
- Number of attention heads $H=16$
- FF-NN projection dimension: 4096
- Head dimension $d_k = d_v = 64$

Write the answer in the box below. Report the results in Millions (M) of parameters, rounded to the nearest integer.

Embedding size: ✓ MTransformer size: ✓ M

For instance,

If the embeddings have 123,456,789 parameters, write **123** (M is implicit).If the transformer layers have 4,500,987,654 parameters, write **4501**.

The described model is Roberta large.

For completeness, the correct number of parameters of roberta large is:

Embedding: 52000768 (51997696 for tokens + position, plus 1 token type embedding (1024) and $1024 * 2$ parameters for layer normalization)Transformer: 302309376 (301989888 from W_q, W_k, W_v, W_o and the two FF-layers, plus $1024 * 2$ (layer norm) and $1024 * 4 + 1024 + 4096$ for the bias terms)**Domanda 7**Risposta
correttaPunteggio
ottenuto 2,00
su 2,00**2 points (no penalty for a wrong answer)**

You are given the following sequence of characters.

a a a b c d d d d c d d

Initially, each character is a separate token.

Apply Byte-Pair Encoding to produce 3 additional tokens. What are the generated tokens?

Write each token, in the correct order, in the following slots.

Separate the merged tokens with a comma (and no spaces).

Examples:

1) if the merged tokens are aa and b, write: **aa,b**2) if the merged tokens are x and y, write: **x,y**1st merge: ✓2nd merge: ✓3rd merge: ✓

Domanda 8

Risposta
errata

Punteggio
ottenuto 0,00
su 1,00

1 point (-25% penalty for each wrong answer)

How does an adversarial agent interaction scheme work?

- ☐ a. The agents work concurrently towards an individual goal, with the objective of maximizing their own benefit.
- ☐ b. The agents work concurrently towards a common goal, with the objective of maximizing the overall benefit.
- ☐ c. The agents work together towards a common goal, in an ordered sequence of subtasks.
- ☐ d. The agents work together towards a common goal, in an unordered sequence of subtasks.
- ☒ e. The agents work independently towards a common goal that is then compared. ✖

Risposta errata.

La risposta corretta è:

The agents work concurrently towards an individual goal, with the objective of maximizing their own benefit.

Commento:

Domanda 9

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-25% penalty for a wrong answer)

The main goal of *instruction tuning* is:

- ☒ a. To fine-tune language models to follow explicit instructions ✔
- ☐ b. To reduce the size of the model without sacrificing performance
- ☐ c. To focus on improving the model's understanding of rare or unseen tokens
- ☐ d. None of the others is the main goal of instruction tuning
- ☐ e. To increase the model's tokenization speed by optimizing the vocabulary size
- ☐ f. To train language models to independently learn new languages from scratch
- ☐ g. To enhance the model's ability to generate longer sequences without truncation

Your answer is correct.

La risposta corretta è:

To fine-tune language models to follow explicit instructions

Domanda 10

Completo

Punteggio
ottenuto 3,00
su 3,00

3 points (no penalty for a wrong answer)

The goal-oriented requirements engineering (GORE) is a technique where the requirements are constructed in three steps:

- user (or agent)
- high-level requirement
- low-level requirement

Consider the case where you receive natural language requirements in a string (called REQS). Describe an architecture of LLAMA agents to perform the steps of the GORE technique, and the prompt engineering techniques employed in each step.

For each agent, write the prompt including System, context, possible inputs from previous prompts, and expected output.

Notice: it is not necessary to write the python calls to the transformers or code for output parsing.

Architecture description:

To extract the three points "user, high and low level requirement" we can use a router chain to route the decision of an agent to three different subchains, each one specialized in one task. One subchains for the user exctraton, one for the high level reqs and one for the low level ones.

Prompt engineering techniques adopted:

Possible techniques adopted could be few-shots prompting.

Prompts:

```
prompt1 = ""
```

You are an expert in extracting user from natural language description of requirements.

```
##Example1:
```

```
*natural language description of a software app regarding physical libriries*
```

```
##Output: Customer, Library Owner, Book, Book supplier
```

```
## Example2:
```

```
*description of an app of online electronic shop.*
```

```
## Output: Shop owner, customer...
```

Now please, with the following information extract the relevant user.

```
{REQS}
```

```
## Output:
```

```
""
```

```
prompt2 = ""
```

You are an expert in extracting high-level requirements from natural language description of requirements.

```
##Example1:
```

```
*natural language description of a software app regarding physical libriries*
```

```
##Output: Customer could buy book, Owner could buy book from the supplier...
```

```
## Example2:
```

```
*description of an app of online electronic shop.*
```

```
## Output: Customer should manage home deliveries of orders.
```

Now please, with the following information extract the relevant high-level requirements.

```
{REQS}
```

```
## Output:
```

```
""
```

```
prompt3 = ""
```

You are an expert in extracting low-level from natural language description of requirements.

##Example1:

natural language description of a software app regarding physical libraries

##Output: Codebase should implement a 'Book' class to model the books owned by the library...

Example2:

description of an app of online electronic shop.

Output: Codebase should implement a 'Address' class to model customers address for deliveries

Now please, with the following information extract the low-level requirements.

{REQS}

Output:

```
""
```

Commento:

**Domanda
11**

Risposta
corretta

Punteggio
ottenuto 2,00
su 2,00

2 points (no penalty for a wrong answer)

The attention for matrices Q, K, V (queries, keys, values) is computed as follows:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$$

Where d_k is the dimensionality of the keys.

The following are the input tokens that reach an attention layer:

$$x = \begin{bmatrix} 3 & 4 & 1 \\ 0 & 4 & 0 \end{bmatrix}$$

The following are the W_q , W_k , W_v matrices for the attention layer:

$$W_q = \begin{bmatrix} -2 & -1 \\ 1 & 1 \\ 0 & -1 \end{bmatrix}$$

$$W_k = \begin{bmatrix} 0 & -2 \\ -1 & 0 \\ -1 & 1 \end{bmatrix}$$

$$W_v = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

Compute the output of the attention operation for the first token.

Write the answer in the following boxes (one dimension for each box).

Report at least 4 significant figures.

Dimension 1: 3,195570314 ✓

Dimension 2: 1,608859364 ✓

Dimension 3: -5.631007774 ✓

**Domanda
12**

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-25% penalty for each wrong answer)

A software repository-based dataset is:

- ☒ a. Any dataset that is based on data extracted from version control systems ✓
- ☐ b. Any dataset that is composed by snippets of code
- ☐ c. Any dataset that is composed by fully working software code artefacts
- ☐ d. Any dataset that is based on open-source code available on online repositories
- ☐ e. Any dataset that is composed by production code artefacts paired with additional information

Risposta corretta.

La risposta corretta è:

Any dataset that is based on data extracted from version control systems